

Pose and Skeleton Representations for Real-Time ASL Recognition

Executive Summary

The automated recognition of American Sign Language (ASL) fundamentally relies on the ability to extract, represent, and sequence the intricate spatio-temporal articulators of the human body. ASL conveys semantic and syntactic information simultaneously through manual parameters—handshape, location, movement, and palm orientation—and non-manual markers (NMMs) such as facial expressions, eye gaze, and head pose¹. Historically, pixel-based deep learning methods have struggled with the severe computational overhead required to process raw RGB video at the high frame rates necessary to capture rapid coarticulation, particularly in fingerspelling tasks. Consequently, lightweight topological representations—specifically, pose and skeleton sequences—have emerged as the foundational modality for real-time ASL recognition¹.

This report provides an exhaustive investigation into the optimal pose- and skeleton-based representations for real-time ASL recognition. Operating under strict real-time computational constraints (sub-200ms latency, ≥ 30 FPS throughput, ≤ 2 GB memory footprint)¹, the analysis systematically evaluates state-of-the-art landmark extractors, prescribes an optimal anatomical landmark set, and defines rigorous mathematical normalizations. Furthermore, it explores the expansive representation search space, contrasting explicitly engineered geometric features with learned representations. The report heavily investigates graph and token-based topologies, detailing strategies for handling corrupted pose data, and critically evaluating the theoretical limitations of pose-only recognition to delineate the exact conditions under which RGB visual features must be retained as a complementary fallback.

Pose-Extraction Technology Comparison

The upstream pose estimator serves as the sensory apparatus for the recognition network. Any error, jitter, or hallucination at this stage acts as adversarial noise to the downstream sequence model. The extraction of hand landmarks, finger joints, wrist locations, upper-body poses, and dense facial features presents a complex multi-task vision problem. Current leading frameworks demonstrate distinct trade-offs between precision, temporal stability, and computational latency.

Extractor Framework	Whole-Body Accuracy	Finger-Detail Quality	Temporal Stability & Occlusion	Latency (Mobile/Edge CPU)	Browser Compatibility	Licensing & Open-Source

MediaPipe Holistic	Low (~50-55 mAP)	Moderate	Poor (High Jitter)	Extremely Fast (<15ms)	Native (WebAssembly)	Apache 2.0 (Google) ⁴
OpenPose	Moderate	Moderate	Moderate	Slow (Requires GPU)	Limited	Non-commercial ⁶
MMPose / RTMW	High (70.2 mAP)	High	Excellent	Fast (ONNX runtime)	Possible via ONNX.js	Apache 2.0 ⁸
DWPose	Very High (77+ mAP)	Very High	Excellent	Moderate	Difficult	Apache 2.0 / Custom ⁶
ViTPose / ZoomNet	Very High	High	High	Slow	No	Research ⁷
Sapiens (Meta)	State-of-the-Art	Flawless (243 face pts)	Exceptional	Very Slow (0.4B-5B params)	No	CC-BY-NC ¹⁰

MediaPipe Holistic: Widely utilized due to its ease of deployment across browsers and mobile devices, MediaPipe operates on a tracking-by-detection paradigm⁴. However, it frequently collapses during rapid ASL motion blur, dropping frames and hallucinating hand topologies when hands cross the torso or face (self-occlusion)⁴. Its temporal jitter destroys the micro-kinematics necessary for analyzing subtle wrist rotations and minimal pairs, making it insufficient for high-fidelity vocabulary scaling.

RTMPose and RTMW (Real-Time Multi-Person Whole-body): Developed by the OpenMMLab team, RTMW targets real-time 2D and 3D whole-body pose estimation. It achieves a groundbreaking 70.2 mAP on the COCO-Wholebody benchmark while maintaining high inference efficiency⁸. By utilizing a coordinate classification mechanism rather than standard heatmap regression, it drastically reduces quantization errors, providing excellent finger-detail quality and occlusion robustness¹².

DWPose (Distillation for Whole-body Pose): DWPose improves upon the RTMPose architecture by employing a two-stage distillation strategy, supervising a student model using a teacher's intermediate features and final logits for both visible and invisible keypoints⁶. This provides exceptional anatomical plausibility even under severe occlusion, making it highly robust for complex ASL configurations where hands interact with the face⁸. However, the architectural overhead slightly increases latency compared to RTMW.

Sapiens / Sapiens2: Meta's recent family of vision transformers sets new benchmarks by natively supporting 1K resolution and featuring up to 5 billion parameters¹⁰. Sapiens extracts 243 facial keypoints alongside highly precise hand kinematics¹⁰. While Sapiens2 achieves state-of-the-art accuracy, its minimum parameter count (400M) violates the strict 2GB memory footprint and 200ms end-to-end latency constraint for edge-deployed real-time systems, restricting its use to server-side processing or offline dataset annotation¹.

Landmark Extractor Recommendation

RTMW (Real-Time Multi-Person Whole-body 3D) is recommended as the primary real-time inference extractor⁸. It provides the necessary >70 mAP accuracy, native 3D lifting capabilities, and operates well within the real-time ONNX CPU/Edge GPU execution budgets¹². **DWPose** is recommended as the offline teacher model for fine-tuning the recognition network's representations and processing training datasets⁶.

Recommended Landmark Set

To preserve the five phonological parameters of ASL, the model must ingest a highly specific subset of whole-body keypoints. Superfluous nodes (e.g., legs and feet) introduce computational noise and increase graph complexity without linguistic benefit, as ASL does not utilize the lower body for semantic generation¹.

The recommended skeletal graph comprises **65 discrete nodes**:

- **Hands (42 nodes):** 21 keypoints per hand (1 wrist, 5 finger bases, 10 intermediate finger joints, 5 fingertips). This dense topology is non-negotiable for distinguishing the approximately 50 distinct ASL handshapes¹⁷.
- **Upper Body (7 nodes):** Nose (1), Shoulders (2), Elbows (2), Wrists (2). These establish the "signing space," absolute spatial reference frame, and macro-kinematic trajectories¹.
- **Facial NMM Regions (16 nodes):** A sparse subset of dense facial meshes designed to capture grammatical non-manual markers. Eyebrows (6 nodes: inner, middle, outer for both eyes to detect raised/furrowed states signaling question types), Eyes (4 nodes: outer/inner corners for gaze trajectory and spatial indexing), Mouth (6 nodes: outer corners, upper/lower lip centers for detecting mouth morphemes like 'cha' or 'th')¹.

Representation Search Space and Mathematical Treatment

Providing a neural network strictly with absolute (x, y, z) coordinates forces it to implicitly learn human anatomy, geometric invariance, and basic Newtonian kinematics before it can learn linguistic syntax. Explicitly engineering spatial and temporal derivatives injects powerful geometric inductive biases, dramatically steepening the learning curve and raising the accuracy ceiling¹. However, the representation search space must balance engineered features against those learned dynamically by the model.

Explicitly engineered features should include coordinate normalizations, bone vectors, and

physical velocities, as these represent deterministic physics. Conversely, learned features should include hand-to-face distances, missing-joint imputations, and multi-scale temporal displacements, as these rely on contextual linguistic intent that is difficult to hardcode.

Let the raw extracted keypoints at time step t be defined as $X_t \in \mathbb{R}^{N \times (C+1)}$, where $N = 65$ (nodes), $C = 3$ (for x, y, z spatial dimensions), and the final channel represents the extractor's confidence score $c \in [0, 1]$. A single node is denoted as $x_{i,t} \in \mathbb{R}^3$.

1. Translation Invariance (Root-Centering)

Absolute camera coordinates are mathematically meaningless because they vary based on the subject's position in the frame. To render the network invariant to where the signer is standing, all coordinates are translated relative to a stable anatomical root (the mid-point of the shoulders).

Equation:

$$x'_{i,t} = x_{i,t} - x_{root,t}$$

Variables:

- $x'_{i,t} \in \mathbb{R}^3$: The translation-normalized spatial coordinate for node i at time t .
- $x_{root,t} = \frac{x_{L_shoulder,t} + x_{R_shoulder,t}}{2} \in \mathbb{R}^3$: The dynamic center of the upper torso.

ASL Representation: This anchors the "signing space" around the upper chest, defining the spatial origin $(0, 0, 0)$. Signs performed at the head possess positive y values, while signs at the waist possess negative y values. It perfectly preserves the linguistic *Place of Articulation* (location parameter)¹. **Failure Cases:** If the pose extractor loses track of a shoulder due to occlusion or severe frame cropping, the root calculation shifts abruptly. This causes the entire constellation of keypoints to jump in the normalized space, triggering catastrophic false-motion artifacts.

2. Signer Scale Normalization

To ensure the network treats a child sitting close to the camera identically to an adult standing far away, the skeleton is scaled by a biologically consistent metric: the bi-acromial (shoulder) breadth.

Equation:

$$x''_{i,t} = \frac{x'_{i,t}}{D_{shoulders,t}}$$

Variables:

- $x''_{i,t} \in \mathbb{R}^3$: The scale-normalized spatial coordinate.

- $D_{shoulders,t} = ||x_{L_shoulder,t} - x_{R_shoulder,t}||_2 \in \mathbb{R}$: The Euclidean distance between the left and right shoulders.

ASL Representation: This ensures that a hand positioned "one shoulder-width away from the chest" is mathematically identical across all users, normalizing the spatial amplitude of the *Movement* parameter.

Failure Cases: When a signer turns in profile (e.g., during ASL role-shifting), the 2D projected shoulder width $D_{shoulders}$ approaches zero. Dividing by a near-zero scalar causes the normalized coordinates to explode toward infinity. This necessitates extracting $D_{shoulders}$ as a median value over the entire sequence, rather than a per-frame calculation, or relying strictly on robust 3D depth-aware distance.

3. Rotation Normalization (Yaw Correction)

For ASL, arbitrary camera angles must be normalized so the signer always appears to face the camera directly.

Equation:

$$x'''_{i,t} = R(\theta, \phi, \psi)x''_{i,t}$$

Variables:

- $R \in \mathbb{R}^{3 \times 3}$: A 3D rotation matrix computed to align the torso's normal vector.
- $v_{shoulders} = (x''_{L_shoulder} - x''_{R_shoulder})$ is rotated to align parallel with the global X-axis, and the vector from the torso center to the nose is aligned with the Y-axis.

ASL Representation: Corrects for a camera placed off-center, standardizing the perspective of palm orientation and movement trajectories toward or away from the body.

Failure Cases: Excessive rotation normalization mathematically scrubs natural ASL role-shifting (where a signer intentionally turns their torso to indicate a change in narrative perspective or character). Careful thresholding must be applied so micro-rotations are corrected, but macro-rotations are preserved.

4. Bone Vector Construction (Spatial Differentials)

Instead of absolute points in space, representations should include the directional vectors of the physical bones.

Equation:

$$b_{i,j,t} = x'''_{j,t} - x'''_{i,t}$$

Variables:

- i, j : Adjacent nodes defined by the anatomical kinematic tree (e.g., wrist to proximal thumb joint).

- $b_{i,j,t} \in \mathbb{R}^3$: The directional bone vector.

ASL Representation: Bone vectors inherently and explicitly encode **Palm Orientation** and **Finger Curvature**. A vector pointing from the wrist to the middle-finger base mathematically defines where the hand is pointing, distinguishing minimal pair signs like "YOUR" (palm facing out) from "MY" (palm facing in) regardless of the absolute location in space¹. **Failure Cases:** Upstream monocular 3D pose estimation is notoriously noisy on the Z-axis (depth). A poorly estimated wrist depth will wildly skew the 3D bone vector orientation, leading to false orientation classifications.

5. Joint Angle Calculation

Equation:

$$\theta_{i,j,k,t} = \arccos \left(\frac{b_{i,j,t} \cdot b_{j,k,t}}{\|b_{i,j,t}\| \cdot \|b_{j,k,t}\|} \right)$$

Variables:

- i, j, k : Sequential joints in a kinetic chain (e.g., proximal, medial, and distal joints of the index finger).
- $\theta_{i,j,k,t} \in \mathbb{R}$: The interior angle between the two adjoining bone vectors.

ASL Representation: Directly encodes **Handshape** (e.g., flat, open palms versus clawed hands) in a manner that is entirely invariant to how the wrist is rotated in 3D space.

Failure Cases: When a finger points directly at the camera lens, foreshortening compresses the 2D projection. If the Z-axis data is weak, the calculated angle will artificially read as 0 degrees (straight), masking bent-finger phonology.

6. Velocity and Acceleration (Temporal Differentials)

Equation (Velocity):

$$v_{i,t} = x'_{i,t} - x'_{i,t-1}$$

Equation (Acceleration):

$$a_{i,t} = v_{i,t} - v_{i,t-1}$$

Variables:

- $v_{i,t}, a_{i,t} \in \mathbb{R}^3$: The first and second temporal derivatives of the spatial coordinates.

ASL Representation: Captures the **Movement** parameter's speed and tension. High acceleration denotes sudden stops, strikes, or bounded movements (e.g., the sharp terminus of the sign "STOP" versus the fluid glide of "SLOW").

Failure Cases: Temporal jitter in the upstream extractor creates massive, artificial spikes in velocity and acceleration, generating high-frequency noise that drowns out actual linguistic

movement.

7. Confidence-Weighted Features

To prevent the model from learning hallucinated geometry caused by occlusion, features must be modulated by the extractor's confidence.

Equation:

$$\tilde{h}_{i,t} = h_{i,t} \odot M(c_{i,t})$$

Variables:

- $h_{i,t} \in \mathbb{R}^F$: The concatenated feature vector for node i (e.g., coordinates, velocity, bone vectors).
- $c_{i,t} \in [0, 1]$: The confidence score provided by the pose extractor.
- $M(c) = c$ (soft weighting) or $M(c) = \mathbb{I}(c > \tau)$ (hard binary masking, where τ is a confidence threshold, typically 0.3).
- $\tilde{h}_{i,t}$: The confidence-weighted feature representation.

ASL Representation: Ensures that when a hand crosses behind the other arm (a common occurrence in two-handed signs), the network ignores the guessing of the pose extractor and instead infers the position through temporal context.

Failure Cases: If the pose extractor is poorly calibrated and outputs uniformly high confidence even during failure, the masking mechanism fails to trigger, corrupting the feature stream.

Graph Construction and Topology

Given the normalized coordinates and derived features, the system must structure the data into a topology that preserves the linguistic syntax of ASL. ASL articulators do not act independently; they exhibit complex, asynchronous coarticulation¹.

Comparison of Representation Architectures

1. **Token-based representations (Transformers):** Flattening the 65 nodes into a sequence of tokens allows standard Attention mechanisms to route information dynamically. While highly expressive, full self-attention scales quadratically ($O(N^2)$) with the number of joints and frames. This violates real-time memory constraints, lacks geometric inductive biases, and ignores the inherent structural sparsity of the human body¹⁹.
2. **Fixed Anatomical Graphs (Standard ST-GCNs):** Nodes are connected solely by physical bones (e.g., wrist to elbow). This fails for ASL because semantic interactions frequently occur across anatomically distant nodes (e.g., the right index finger physically touching the chin in the sign "MOTHER"). Information must travel an inefficiently long path through the arms and neck to relate the finger to the face¹.
3. **Heterogeneous Multi-scale Graphs:** Assigns different semantic types to different

nodes (e.g., classifying nodes as 'hand', 'face', or 'body'). This allows relation-specific edge types, preventing the graph from processing a delicate eyelid movement with the same convolutional weights used for a massive shoulder swing.

4. **Spatio-Temporal Hypergraphs:** Unlike standard graphs where edges connect exactly two nodes, a hyperedge can connect an arbitrary set of nodes²⁰. A single hyperedge can encompass all five fingertips, capturing global hand states (like 'open palm' or 'fist') without requiring message passing through the intermediate knuckles.

Recommended Topology: Dynamic Heterogeneous Graph

The optimal representation utilizes a Spatio-Temporal Graph Convolutional Network (ST-GCN) backbone enhanced with relation-specific edge types and dynamic cross-attention. This mirrors the high-efficiency architecture of models like SignBart, which separately encodes X and Y coordinates to prevent spatial collapse and achieves high accuracy with sub-10 million parameters³.

8. Graph Adjacency Construction

Let the graph be defined as $G = (V, E)$, with an Adjacency Matrix $A \in \mathbb{R}^{N \times N}$. The node feature tensor for a single frame is $H \in \mathbb{R}^{N \times F}$ (where F is the concatenation of coordinates, bone vectors, and velocities).

Equation:

$$H^{(l+1)} = \sigma \left(\sum_{k=1}^K A_k H^{(l)} W_k^{(l)} \right)$$

Variables:

- $H^{(l)}$: The node representations at layer l .
- $A_k \in \mathbb{R}^{N \times N}$: A set of K distinct, fixed adjacency matrices (e.g., inward-pointing anatomical edges, outward-pointing edges, self-loops).
- $W_k^{(l)} \in \mathbb{R}^{F_{in} \times F_{out}}$: Learnable weight matrices for edge type k .
- σ : Non-linear activation (e.g., ReLU).

ASL Representation: This base formulation allows the network to process the physical structure of the signer, distributing information along the bones.

Failure Cases: Fails to capture non-physical semantic relations (like hand-to-face proximity).

9. Dynamic Graph Construction

To model interactions between disconnected body parts, the adjacency matrix must adapt based on the data.

Equation:

$$A_{dynamic,t} = \text{Softmax} \left(\frac{(H_t W_Q)(H_t W_K)^T}{\sqrt{d_k}} \right)$$

Variables:

- $A_{dynamic,t} \in \mathbb{R}^{N \times N}$: The data-dependent adjacency matrix at time t .
- $W_Q, W_K \in \mathbb{R}^{F \times d_k}$: Learnable projection matrices generating queries and keys.

ASL Representation: This attention-derived graph allows the network to dynamically create strong edges when the dominant hand moves near the non-dominant hand or the face,

directly mapping the *Location* phonology of ASL²³. **Failure Cases:** If $A_{dynamic,t}$ is not aggressively regularized or made sparse (e.g., via K-Nearest-Neighbors pruning), the graph becomes fully connected. This causes over-smoothing, blending all high-frequency spatial details and destroying the distinction between subtle minimal pairs²⁴.

10. Temporal Graph Construction

Nodes must connect not just spatially, but across time.

Equation:

$$H_{out} = \text{Conv1D}_{temporal}(H_{spatial_out})$$

Variables:

- $H_{spatial_out} \in \mathbb{R}^{T \times N \times F'}$: The output of the spatial GCN across the temporal window T .
- $\text{Conv1D}_{temporal}$: A 1D temporal convolution applied across the time dimension for each node independently, with a kernel size K_t .

ASL Representation: Captures the trajectory and coarticulation of the signs. It tracks how a single joint (e.g., the wrist) moves over a half-second window.

Failure Cases: If K_t is too large, the network blurs sequential signs together; if too small, it fails to capture the macro-movement parameter.

Missing and Noisy Pose Data Strategy

Self-occlusion is a physical certainty in ASL. When a hand crosses the torso or face, upstream extractors (even robust ones like DWPose/RTMW) will degrade, exhibiting temporal jitter, missing landmarks, or left-right hand swaps¹⁴. The downstream model must be mathematically resilient through specialized data processing.

11. High-Frequency Jitter: One Euro Filtering

To mitigate micro-tremors in the pose extractor without introducing the severe latency lag of a

standard moving average or Kalman filter, the system must implement the **One Euro Filter**²⁷. It acts as a first-order low-pass filter with an adaptive cutoff frequency.

Equation:

$$X_{f,t} = \alpha X_t + (1 - \alpha) X_{f,t-1}$$

Variables:

- $X_{f,t} \in \mathbb{R}^3$: The filtered coordinate.
- $\alpha = \frac{1}{1 + \frac{\tau_c}{T_e}} \in [0, 1]$: The smoothing factor. T_e is the sampling period (1/FPS).
- $\tau_c = \frac{1}{2\pi f_c}$, where the cutoff frequency $f_c = f_{min} + \beta|v_t|$.
- f_{min}, β : Tunable parameters controlling minimum cutoff and speed coefficient.

ASL Representation: During slow motions (e.g., a static handshake hold), velocity $|v_t|$ is low, resulting in a low cutoff frequency. The filter heavily smooths the data, locking the hand in place and eliminating jitter. During rapid fingerspelling, velocity is high, instantly raising the cutoff

frequency to eliminate latency lag²⁷. **Failure Cases:** If β is tuned too low, rapid fingerspelling micro-movements are interpreted as noise and scrubbed, destroying the temporal coarticulation necessary for character recognition.

Imputation and Masked Modeling

Beyond filtering, missing nodes (flagged by confidence thresholding) require active recovery:

- **Temporal Interpolation:** Linear or spline interpolation is acceptable for gaps of 1-3 frames. However, ASL motion is highly non-linear; interpolating a missing hand over 10 frames of a ballistic movement will generate anatomically impossible trajectories.
- **Masked Pose Modeling (Learned Imputation):** By applying modality dropout during training (intentionally zeroing out hand coordinates 20% of the time), the GCN learns to implicitly impute the location of the missing hand based on the trajectory of the elbow and shoulder, mirroring self-supervised masked autoencoder strategies.
- **Kalman Filtering:** While useful for linear trajectory prediction, Kalman filters often fail in ASL because hand movements are characterized by sudden, unpredictable directional changes (e.g., zig-zag motions) that violate linear state assumptions.

Pose-Only Feasibility and RGB Fallback

Critically, **pose alone is mathematically insufficient** for mapping the entirety of ASL semantics. Relying exclusively on skeletal graphs establishes an unavoidable asymptotic accuracy ceiling due to inherent information destruction¹.

Limitations of Pose-Only Representations

1. **Fine Finger Contact & Thumb Placement:** Extractors like RTMW and Sapiens represent the thumb with only 4 topological keypoints. This wireframe topology lacks physical

volume. Distinguishing between the ASL letter 'M' (thumb tucked under three fingers) and 'N' (thumb tucked under two) requires tracking exact 3D flesh contact and micro-occlusions¹. Pose estimators frequently miscalculate these states, generating identical coordinate outputs for different semantic letters.

2. **Visually Similar Signs:** Minimal pairs separated only by subtle cheek contact or micro-tension cannot be resolved via skeletons.
3. **Facial Grammar and Mouth Shape:** Sparse facial keypoints (e.g., 6 points on the lips) cannot capture complex facial muscle tensions, tongue positions (e.g., the 'th' morpheme indicating carelessness), or subtle cheek puffing, completely scrubbing this grammatical layer from the data.
4. **Subtle Wrist Rotation (Radioulnar Articulation):** The orientation of the palm is inferred from the 3D projection of the wrist-to-knuckle vector. In monocular video, depth ambiguity frequently destroys the palm orientation parameter⁴.

Conditions for RGB Hand Crop Injection

To maintain the $< 200ms$ latency constraint, the system cannot process full-frame RGB continuously. Instead, an auxiliary RGB perception stream must be triggered conditionally:

- **Fingerspelling Trigger:** When the macro-kinematic trajectory of the wrist halts but internal finger velocity remains high (the classical spatial signature of fingerspelling), the system must explicitly extract a 64×64 dynamic RGB bounding-box crop of the dominant hand³.
- **Confidence Collapse:** If the pose extractor's average hand confidence drops below 0.4 (indicating severe self-occlusion or motion blur), the RGB crop is passed to a lightweight ResNet or MobileNet block (e.g., the HandReader architecture) to extract localized pixel-based textures to complement the failing topological graph²⁹.

Final Recommendation: Three Representation Candidates

Based on the synthesis of latency requirements, linguistic phonology, and state-of-the-art architectures, three candidates are presented:

1. The Simple Baseline: MediaPipe + Absolute Coordinates + LSTM

- **Architecture:** MediaPipe Holistic $\rightarrow$ Absolute 2D (x, y, c) coordinates $\rightarrow$ Flattened token vector $\rightarrow$ Bi-directional LSTM.
- **Verdict:** Highly efficient, easily deployed via WebAssembly in browsers. **However**, it will catastrophically fail at ISLR vocabulary scaling due to a lack of spatial invariance (failing when subjects move off-center) and an inability to model non-linear spatial coarticulation. It cannot serve a production environment.

2. The Strong Evidence-Backed Candidate: RTMW + Body-Normalized

+ ST-GCN

- **Architecture:** RTMW (ONNX) $\rightarrow$ Body-Normalized 3D Coordinates + Bone Vectors + Confidence Masking $\rightarrow$ Separated X/Y encoding $\rightarrow$ Spatio-Temporal Graph Convolution (SignBart style).
- **Verdict:** This is the **optimal production-ready choice**. By normalizing against the torso, extracting explicit bone vectors, and routing data through an ST-GCN (which requires exponentially fewer FLOPs than a standard vision transformer), the system achieves deep linguistic fidelity. It operates well within the 2GB memory and 200ms latency budgets while preserving minimal pairs¹.

3. The Novel/Underexplored Candidate: 3D Lifted Poses + Hypergraphs + State Space Models (Mamba)

- **Architecture:** Multi-view 3D lifted pose $\rightarrow$ Spatio-Temporal Hypergraph Convolution (ST-HCSS) to aggregate whole-hand/whole-body states $\rightarrow$ State Space Model (SSM / Mamba).
- **Verdict:** Extremely promising for unsegmented continuous recognition. SSMs (like Mamba)³⁰ avoid the quadratic scaling of attention, allowing infinite context windows with $O(1)$ inference latency. Hypergraphs elegantly handle the compositional nature of ASL phonology (e.g., linking all fingers simultaneously)²⁰. However, this poses severe implementation risks within a standard development sprint due to the immaturity of hypergraph-SSM toolchains.

Recommended Real-Time Preprocessing Pipeline & Pseudocode

To execute the Strong Evidence-Backed representation in real-time, the pipeline must operate via a stateful streaming buffer, avoiding recalculation of past frames.

Pipeline Steps:

1. **Ingestion:** Capture RGB frame at 30 FPS.
2. **Perception:** Pass through RTMW (ONNX). Yields $X_{raw} \in \mathbb{R}^{65 \times 4}$ (x, y, z, c).
3. **Filtering:** Apply One Euro Filter to update $X_{filtered}$ in-place using previous state buffers to kill jitter.
4. **Normalization:** Calculate shoulder root. Subtract root. Divide by median shoulder width.
5. **Derivation:** Compute velocity v and bone vectors b .
6. **Concatenation:** Form the node feature matrix $H_t = [X_{\{norm\}} \parallel \vec{v}]$

$\|\vec{b}\|$
 $\|\vec{c}\|$.

7. **Buffer:** Append H_t to a sliding window buffer $W \in \mathbb{R}^{T \times 65 \times 10}$ (where $T = 32$ frames).

8. **Inference:** Pass W to the ST-GCN embedding projector once the window is full or a boundary is detected.

Implementation Pseudocode

Python

```
class ASLPreprocessor:
```

```
    def __init__(self, one_euro_params):
```

```
        self.one_euro = OneEuroFilter(*one_euro_params)
```

```
        self.pose_extractor = RTMWEONNX_Engine()
```

```
        self.buffer = collections.deque(maxlen=32)
```

```
        self.prev_x = None
```

```
    def process_frame(self, frame_rgb):
```

```
        # 1. Extraction (x, y, z, conf)
```

```
        raw_pose = self.pose_extractor.infer(frame_rgb)
```

```
        # 2. Filtering (High-frequency jitter removal)
```

```
        smooth_pose = self.one_euro.process(raw_pose)
```

```
        # 3. Normalization (Translation & Scale)
```

```
        l_sh, r_sh = smooth_pose[SHOULDER_L], smooth_pose[SHOULDER_R]
```

```
        root = (l_sh + r_sh) / 2.0
```

```
        scale = torch.linalg.norm(l_sh - r_sh) + 1e-6
```

```
        norm_pose = (smooth_pose - root) / scale
```

```
        # 4. Derivations
```

```
        velocity = (norm_pose - self.prev_x if self.prev_x else torch.zeros_like(norm_pose))
```

```
        bone_vectors = self.compute_bones(norm_pose)
```

```
        self.prev_x = norm_pose
```

```
        # 5. Feature Concatenation & Confidence Masking
```

```
        confidence = smooth_pose[3:4]
```

```
        # Shape: (65 nodes, 10 features)
```

```

node_features = torch.cat([norm_pose[-3], velocity[-3], bone_vectors, confidence],
                           dim=-1)

# Hard masking for nodes with low confidence
node_features = node_features * (confidence > 0.3)

self.buffer.append(node_features)

if len(self.buffer) == 32:
    return torch.stack(list(self.buffer)) # Yields (32, 65, 10)
return None

```

Research Gaps

1. **3D Depth Estimation from Monocular 2D:** While models like RTMW attempt 3D lifting, extracting accurate depth (Z-axis) of the hands relative to the torso from standard 2D webcams remains highly unstable. This depth ambiguity destroys the *Location* parameter when hands move directly toward the lens rather than laterally.
2. **Dataset Bias in Normalization:** Normalizing by shoulder width assumes standard body morphologies. It risks failing on signers with physical deformities, extreme obesity, or those seated in wheelchairs, presenting an unexplored fairness and accessibility gap¹.
3. **Temporal Boundaries of Fingerspelling:** Real-time switching between full-body ST-GCN processing and localized RGB sequence-to-sequence decoding (ByT5) requires mathematically precise temporal boundary detection (sign spotting) to know exactly *when* fingerspelling begins and ends¹.

Handoff to the Next Research Stage

This document specifies the exact topological extraction, geometric normalization, and feature derivation requirements. The engineering and architecture teams must adhere to the following input specifications to begin spatial architecture research.

Exact Input Tensor Specification

- **Data Structure:** Spatio-Temporal Graph Feature Matrix.
- **Shape:** (Batch, Time, Nodes, Features) $\rightarrow$ (B, T, 65, 10).
 - Time (T): Dynamic sequence length (padded) or fixed sliding window (e.g., $T = 32$ to 64 frames depending on sample rate).
 - Nodes (V): 65 (42 hand nodes, 7 upper body nodes, 16 facial nodes).
 - Features (C): 10 channels per node.
 - Channels 0-2: Body-normalized 3D spatial coordinates (x, y, z) .

- Channels 3-5: Instantaneous 3D velocity vectors (v_x, v_y, v_z) .
- Channels 6-8: 3D Bone vectors originating from the node (b_x, b_y, b_z) .
- Channel 9: Confidence score $c \in [0.0, 1.0]$.

Node Types, Edge Types, and Normalization

- **Nodes:** Explicitly segregate the graph into three heterogeneous node sub-types (Hand, Body, Face) to allow relation-specific embedding layers before GCN propagation.
- **Edges:** Incorporate fixed anatomical edges, fixed left-right symmetric cross-body edges, and learned dynamic attention edges to capture hand-to-face proximity.
- **Normalization Procedure:** Translate the origin to the shoulder mid-point. Scale the entire skeleton by the absolute shoulder width.
- **Missing-Data Masks:** Nodes with confidence < 0.3 must have their features masked to 0.0 , forcing the downstream GCN to impute the position via temporal attention rather than processing hallucinated noise.

Requirements for Spatial Architecture Research

- **Pose-Only Limitations:** The downstream architecture **must** contain an auxiliary routing mechanism to accept cropped RGB tensors (B, T, 3, 64, 64) of the dominant hand. This stream should only activate via a gating mechanism when pose confidence collapses or during explicit fingerspelling CTC decoding.
- **Landmark Extractor Recommendation:** Implement using **MMPose RTMW** (ONNX deployment format) for the optimal balance of 70+ mAP accuracy and real-time CPU/Edge GPU inference.

Top 20 Relevant Sources

1. *ASL Citizen: A Community-Sourced Dataset for Advancing Isolated Sign Language Recognition* (Desai et al.)¹
2. *RTMW: Real-Time Multi-Person 2D and 3D Whole-body Pose Estimation* (Jiang et al.)⁸
3. *Effective Whole-body Pose Estimation with Two-stages Distillation (DWPose)* (Yang et al.)⁶
4. *Sapiens: Foundation for Human Vision Models* (Meta AI)¹⁰
5. *SignBart: New approach with the skeleton sequence for Isolated Sign language Recognition* (Nguyen et al.)³
6. *Improving Handshape Representations for Sign Language Processing: A Graph Neural Network Approach* (Carbo & Nalisnick)¹⁷
7. *1 € Filter: A Simple Speed-based Low-pass Filter for Noisy Input in Interactive Systems* (Casiez et al.)²⁷
8. *State Space Models are Effective Sign Language Learners: Exploiting Phonological Compositionality* (Cheng et al.)³⁰

9. *HandReader: Advanced Techniques for Efficient Fingerspelling Recognition* (Korotaev et al.)²⁹
10. *Spatio-Temporal Hypergraph Convolutional Neural Network for Soft Sensing* (Tew et al.)²⁰
11. *PoseBH: Prototypical Multi-Dataset Training Beyond Human Pose*
[cite: 33]
12. *ZoomNAS: Searching for Whole-body Human Pose Estimation in the Wild*
[cite: 7]
13. *Long-Term Spatio-Temporal Graph Attention Network for Traffic Forecasting*
[cite: 23]
14. *Sign Language Recognition in the Age of LLMs*
[cite: 1]
15. *Fingerspelling within Sign Language Translation*
[cite: 1]
16. *Phonological Perception of Sign Language Models*
[cite: 1]
17. *One-Shot Learning for Isolated Sign Language Recognition across Languages*
[cite: 1]
18. *American Sign Language Handshapes Reflect Pressures for Communicative Efficiency*
[cite: 1]
19. *The Sem-Lex Benchmark: Modeling ASL Signs and Their Phonemes*
[cite: 1]
20. *Isolated Sign Language Recognition with Segmentation and Pose Estimation*
[cite: 1]

Works cited

1. Real-Time ASL Recognition Specification.pdf
2. Phonological Representation Learning for Isolated Signs Improves Out-of-Vocabulary Generalization - arXiv, <https://arxiv.org/html/2509.04745v1>
3. SignBart - New Approach with the Skeleton Sequence for Isolated Sign Language Recognition | Request PDF - ResearchGate, https://www.researchgate.net/publication/397435992_SignBart_-_New_Approach_with_the_Skeleton_Sequence_for_Isolated_Sign_Language_Recognition
4. Whole-Body Human Pose Estimation in the Wild | Request PDF - ResearchGate, https://www.researchgate.net/publication/346667460_Whole-Body_Human_Pose_Estimation_in_the_Wild
5. Awesome Computer Vision Resources: From Foundations to Research - GitHub, <https://github.com/mawady/awesome-cv>
6. Single-Network Whole-Body Pose Estimation | Request PDF - ResearchGate, https://www.researchgate.net/publication/339560581_Single-Network_Whole-Body_Pose_Estimation
7. ZoomNAS: Searching for Whole-body Human Pose Estimation in the Wild - ResearchGate, https://www.researchgate.net/publication/362575778_ZoomNAS_Searching_for_

- [Whole-body Human Pose Estimation in the Wild](#)
8. 2024_RTMW: Real-Time Multi-Person 2D and 3D Whole-body Pose Estimation - Scribd,
<https://www.scribd.com/document/1065595357/2024-RTMW-Real-Time-Multi-Person-2D-and-3D-Whole-body-Pose-Estimation>
 9. Human Scene Reconstruction for Urban Search and Rescue Missions - reposiTUm,
<https://repositum.tuwien.at/bitstream/20.500.12708/221867/1/Purgat%20Martin%20-%202025%20-%20Human%20Scene%20Reconstruction%20for%20Urban%20Search%20and%20Rescue...pdf>
 10. Meta's Sapiens: Revolutionizing Human Pose, Segmentation, and Depth Estimation with Vision Transformers - Synced Review,
<https://syncedreview.com/2024/08/27/metaspaiens-revolutionizing-human-pose-segmentation-and-depth-estimation-with-vision-transformers/>
 11. Sapiens2 - arXiv, <https://arxiv.org/html/2604.21681v1>
 12. Lightweight Facial Landmark Detection in Thermal Images via Multi-Level Cross-Modal Knowledge Transfer - arXiv, <https://arxiv.org/html/2510.11128v1>
 13. BRNO UNIVERSITY OF TECHNOLOGY APP SUPPORTING STRENGTH SPORTS TRAINING - IT SPY,
https://www.itspy.cz/wp-content/uploads/2024/09/it_spy_2024_diplomova_prace_51.pdf
 14. Effective Whole-body Pose Estimation with Two-stages Distillation - ResearchGate,
https://www.researchgate.net/publication/376827201_Effective_Whole-body_Pose_Estimation_with_Two-stages_Distillation
 15. SAPIENS2 - ICLR Proceedings - ICLR 2026,
https://proceedings.iclr.cc/paper_files/paper/2026/file/5fd68e1c262099f846733435d620d574-Paper-Conference.pdf
 16. Naomi Caselli's research works | Boston University and other places - ResearchGate,
<https://www.researchgate.net/scientific-contributions/Naoimi-Caselli-2161786801>
 17. Improving Handshape Representations for Sign Language Processing: A Graph Neural Network Approach - ACL Anthology,
<https://aclanthology.org/2025.emnlp-main.1483/>
 18. Natural Language-Assisted Sign Language Recognition | Request PDF - ResearchGate,
https://www.researchgate.net/publication/373323643_Natural_Language-Assisted_Sign_Language_Recognition
 19. Sapiens2 | OpenReview, <https://openreview.net/forum?id=IVAIYCqdvW>
 20. Hwa Hui Tew - CatalyzeX, <https://www.catalyzex.com/author/Hwa%20Hui%20Tew>
 21. UPTRec: Fusing User Graph, Point-of-Interest Transitions, and Temporal Embeddings for Next Point-of-Interest Recommendations - MDPI,
<https://www.mdpi.com/2220-9964/15/3/122>
 22. (PDF) Sign Language Recognition – Dataset Cleaning for Robust Word Classification in a Landmark-Based Approach - ResearchGate,

- https://www.researchgate.net/publication/391380947_Sign_Language_Recognition_-_Dataset_Cleaning_for_Robust_Word_Classification_in_a_Landmark-Based_Approach
23. Long-Term Spatio-Temporal Graph Attention Network for Traffic Forecasting - ResearchGate,
https://www.researchgate.net/publication/391948098_Long-Term_Spatio-Temporal_Graph_Attention_Network_for_Traffic_Forecasting
 24. Responsible AI-Enabled Infodemic Management: A Hypergraph-Based Infodemic Topic Prediction Framework | INFORMS Journal on Computing - PubsOnLine,
<https://pubsonline.informs.org/doi/10.1287/ijoc.2024.0660>
 25. Skeleton Aware Multi-modal Sign Language Recognition | Request PDF - ResearchGate,
https://www.researchgate.net/publication/354297024_Skeleton_Aware_Multi-modal_Sign_Language_Recognition
 26. SmoothNet: A Plug-and-Play Network for Refining Human Poses in Videos - ECVA,
https://www.ecva.net/papers/eccv_2022/papers_ECCV/papers/136650615.pdf
 27. 1€ Filter: A Simple Speed-based Low-pass Filter for Noisy Input in Interactive Systems,
https://www.researchgate.net/publication/254005010_1_Filter_A_Simple_Speed-based_Low-pass_Filter_for_Noisy_Input_in_Interactive_Systems
 28. Contagious infection-free medical interaction system with machine vision controlled by remote hand gesture during an operation - PMC,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC11127465/>
 29. [2505.10267] HandReader: Advanced Techniques for Efficient Fingerspelling Recognition, <https://arxiv.org/abs/2505.10267>
 30. State Space Models are Effective Sign Language Learners: Exploiting Phonological Compositionality for Vocabulary-Scale Recognition - arXiv,
<https://arxiv.org/html/2604.08761v1>
 31. One-Stage 3D Whole-Body Mesh Recovery with Component Aware Transformer | Request PDF - ResearchGate,
https://www.researchgate.net/publication/373324847_One-Stage_3D_Whole-Body_Mesh_Recovery_with_Component_Aware_Transformer
 32. Alessa Carbo, <https://alessac14.github.io/>
 33. CVPR Poster PoseBH: Prototypical Multi-Dataset Training Beyond Human Pose Estimation, <https://cvpr.thecvf.com/virtual/2025/poster/33228>